DisasterAware - rules the system wrote and changed

Source: dss_generated_state.json - 81 written, 12 tuned

Every rule below was produced or changed during a run. The seed rule base is not listed here: those rules were written by hand before any fire started and belong to their own appendix. What follows is what the adaptation stages did to that base. The resolution stage instantiates a rule when the situation in front of it falls on an antecedent cell the standing base does not answer, building it from the nearest standing rule. The generative stage proposes a rule outright, and nothing it proposes acts before it has passed the full gate chain. The evolving-fuzzy stage does not add rules; it walks the consequents of rules that already exist, one accepted step at a time, and a step is kept only when the reseeded forecast shows it costs less than the standing decision. Each row carries the step and minute at which the change was made and the record it came from, so any line here can be traced back to the run that produced it.

Summary

What the system did	Rules
Wrote a rule outright, generative stage (Stage 3)	40
Instantiated a rule on an unanswered antecedent cell, resolution stage (Stage 2)	41
Tuned the consequents of an existing rule, evolving-fuzzy stage (Stage 1)	12
Reshaped an input variable (boundary moved or term inserted)	9

Reading the rules

Symbol	Reading
VL / L / M / H / VH	very low, low, moderate, high, very high: the five-term partition every input variable carries.
>=H, >=M, >=L	at least that term: the rule fires on the term and everything above it.
X1, X2, X3	terms the resolution stage inserted into a variable when the standing five were too coarse to separate the situation it was facing. They sit between the standing terms and preserve the partition.
0.00 - 1.00	the consequent value: how hard the named action is ordered, on the scale every rule shares.

Rules the system wrote

ID	Rule	Description	Origin	Produced
G1	IF fire threat level is H AND suppression feasibility is VH THEN suppression effort 0.90, retardant drop 0.80, resource deployment 0.70	When the fire threat is high and suppression feasibility is very high, press the suppression effort at full strength (0.90), drop retardant hard (0.80), deploy resources hard (0.70).	Generative (Stage 3)	step 9, 5 min, deficit 0.2 gen_rule_0001
G2	IF suppression feasibility is VH AND intervention urgency is VH THEN suppression effort 0.90, retardant drop 0.80, resource deployment 0.70	When suppression feasibility is very high and intervention urgency is very high, press the suppression effort at full strength (0.90), drop retardant hard (0.80), deploy resources hard (0.70).	Generative (Stage 3)	step 16, 11 min, deficit 0.08 gen_rule_0003
G3	IF suppression feasibility is VH AND fire threat level is H AND intervention urgency is VH THEN suppression effort 0.95, retardant drop 0.85, resource deployment 0.90	When suppression feasibility is very high and the fire threat is high and intervention urgency is very high, press the suppression effort at full strength (0.95), drop retardant at full strength (0.85), deploy resources at full strength (0.90).	Generative (Stage 3)	step 10, 10 min, deficit 0.18 gen_rule_0004
G4	IF suppression feasibility is VH AND intervention urgency is VH AND fire threat level is M THEN	When suppression feasibility is very high and intervention urgency is very high and the fire threat is moderate,	Generative (Stage 3)	step 25, 25 min, deficit 0.3 gen_rule_0006

	downwind backburn 0.90, retardant drop 0.80, asset protection 0.60	order downwind backburn at full strength (0.90), drop retardant hard (0.80), protect the exposed assets moderately (0.60). "downwind backburn" is a generated intervention (containment line x1.0 + suppression effort x0.7).		
G5	IF asset exposure risk is VH AND suppression feasibility is VH THEN downwind retardant shield 0.90, asset protection 0.70, resource deployment 0.70	When asset exposure risk is very high and suppression feasibility is very high, order downwind retardant shield at full strength (0.90), protect the exposed assets hard (0.70), deploy resources hard (0.70). "downwind retardant shield" is a generated intervention (retardant drop x1.0 + containment line x0.8).	Generative (Stage 3)	step 5, 5 min, deficit 0.12 gen_rule_0008
G6	IF suppression feasibility is VH AND fire threat level is >=H THEN drafting sustained attack 0.95, resource deployment 0.80	When suppression feasibility is very high and the fire threat is at least high, order drafting sustained attack at full strength (0.95), deploy resources hard (0.80). "drafting sustained attack" is a generated intervention (water drafting x1.0 + suppression effort x0.9).	Generative (Stage 3)	step 10, 10 min, deficit 0.1 gen_rule_0010
G7	IF fire threat level is >=H AND suppression feasibility is VH THEN head knockdown 0.95, resource deployment 0.80	When the fire threat is at least high and suppression feasibility is very high, order head knockdown at full strength (0.95), deploy resources hard (0.80). "head knockdown" is a generated intervention (water drafting x1.0 + suppression effort x0.9 + retardant drop x0.6).	Generative (Stage 3)	step 10, 10 min, deficit 0.08 gen_rule_0013
G8	IF suppression feasibility is VH AND fire threat level is >=H AND evacuation pressure is VH THEN wet containment line 0.90, resource deployment 0.80	When suppression feasibility is very high and the fire threat is at least high and evacuation pressure is very high, order wet containment line at full strength (0.90), deploy resources hard (0.80). "wet containment line" is a generated intervention (water drafting x1.0 + containment line x0.8).	Generative (Stage 3)	step 15, 15 min, deficit 0.16 gen_rule_0015
G9	IF suppression feasibility is VH AND fire threat level is H THEN counterfire strip 0.90, resource deployment 0.80	When suppression feasibility is very high and the fire threat is high, order counterfire strip at full strength (0.90), deploy resources hard (0.80). "counterfire strip" is a generated intervention (tactical burn x1.0 + containment line x0.6).	Generative (Stage 3)	step 27, 13 min, deficit 0.05 gen_rule_0020
G12	IF suppression feasibility is VH AND asset exposure risk is VH AND fire threat level is VL THEN suppression effort 0.90, water drafting 0.80, resource deployment 0.80	When suppression feasibility is very high and asset exposure risk is very high and the fire threat is very low, press the suppression effort at full strength (0.90), draft water hard (0.80), deploy resources hard (0.80). renamed from G9 (name collision across sessions).	Generative (Stage 3)	step 5, 5 min, deficit 0.1 gen_rule_0021
G10	IF suppression feasibility is VH AND asset exposure risk is H AND evacuation pressure is H	When suppression feasibility is very high and asset exposure risk is high	Generative (Stage 3)	step 35, 35 min, deficit 0.06 gen_rule_0024

	THEN suppression effort 0.90, water drafting 0.80, resource deployment 0.70	and evacuation pressure is high, press the suppression effort at full strength (0.90), draft water hard (0.80), deploy resources hard (0.70).		
G11	IF asset exposure risk is VH AND suppression feasibility is VH AND intervention urgency is VL THEN asset protection 0.90, containment line 0.80, retardant drop 0.70	When asset exposure risk is very high and suppression feasibility is very high and intervention urgency is very low, protect the exposed assets at full strength (0.90), cut containment line hard (0.80), drop retardant hard (0.70).	Generative (Stage 3)	step 29, 0 min, deficit 0.005000000000000003 gen_rule_0035
G13	IF interface exposure is VH AND suppression feasibility is VH THEN asset protection 0.90, containment line 0.80, resource deployment 0.70	When interface exposure is very high and suppression feasibility is very high, protect the exposed assets at full strength (0.90), cut containment line hard (0.80), deploy resources hard (0.70). "interface exposure" is a generated concept (asset exposure risk x0.5 + evacuation pressure x0.5).	Generative (Stage 3)	step 15, 5 min, deficit 0.3 gen_rule_0061
G14	IF suppression feasibility is VH AND evacuation pressure is VH AND fire threat level is VL THEN suppression effort 0.90, water drafting 0.80, resource deployment 0.80	When suppression feasibility is very high and evacuation pressure is very high and the fire threat is very low, press the suppression effort at full strength (0.90), draft water hard (0.80), deploy resources hard (0.80). Written 2 times over the recorded runs.	Generative (Stage 3)	step 11, 6 min, deficit 0.3 gen_rule_0062
G16	IF suppression feasibility is VH AND asset exposure risk is VH AND evacuation pressure is VH THEN suppression effort 0.90, water drafting 0.80, retardant drop 0.70	When suppression feasibility is very high and asset exposure risk is very high and evacuation pressure is very high, press the suppression effort at full strength (0.90), draft water hard (0.80), drop retardant hard (0.70).	Generative (Stage 3)	step 6, 6 min, deficit 0.3 gen_rule_0066
G17	IF suppression feasibility is VH AND asset exposure risk is VH AND intervention urgency is VH THEN suppression effort 0.95, water drafting 0.80, retardant drop 0.70	When suppression feasibility is very high and asset exposure risk is very high and intervention urgency is very high, press the suppression effort at full strength (0.95), draft water hard (0.80), drop retardant hard (0.70).	Generative (Stage 3)	step 51, 31 min, deficit 0.2 gen_rule_0069
G18	IF suppression feasibility is M AND asset exposure risk is VH AND evacuation pressure is VH THEN suppression effort 0.90, water drafting 0.80, retardant drop 0.70	When suppression feasibility is moderate and asset exposure risk is very high and evacuation pressure is very high, press the suppression effort at full strength (0.90), draft water hard (0.80), drop retardant hard (0.70).	Generative (Stage 3)	step 83, 83 min, deficit 0.18 gen_rule_0088
G19	IF fire threat level is H AND asset exposure risk is VH AND suppression feasibility is VH THEN suppression effort 0.90, retardant drop 0.80, asset protection 0.70	When the fire threat is high and asset exposure risk is very high and suppression feasibility is very high, press the suppression effort at full strength (0.90), drop retardant hard (0.80), protect the exposed assets hard (0.70).	Generative (Stage 3)	step 6, 0 min, deficit 0.005000000000000001 gen_rule_0122
G20	IF asset exposure risk is VH AND suppression feasibility is VH AND evacuation pressure is VL THEN containment line 0.80,	When asset exposure risk is very high and suppression feasibility is very high and evacuation pressure is very low, cut containment line hard (0.80), drop	Generative (Stage 3)	step 6, 0 min, deficit 0.005000000000000003 gen_rule_0127

	retardant drop 0.70, resource deployment 0.70	retardant hard (0.70), deploy resources hard (0.70).		
G21	IF fire threat level is H AND suppression feasibility is L THEN containment line 0.90, retardant drop 0.80	When the fire threat is high and suppression feasibility is low, cut containment line at full strength (0.90), drop retardant hard (0.80). Recorded reason: fills the pre-VH gap G1-G4 leave; this fire lived at H/low-feasibility where only defensive rules fired.	Generative (Stage 3)	gen_rule_0133
G22	IF fire threat level is >=H AND suppression feasibility is L THEN retardant drop 0.90, containment line 0.80	When the fire threat is at least high and suppression feasibility is low, drop retardant at full strength (0.90), cut containment line hard (0.80). Recorded reason: Direct attack was infeasible yet aerial retardant needs no roads and the map has water to refill, the only path to bend the head here.	Generative (Stage 3)	gen_rule_0135
G23	IF suppression feasibility is VH AND fire threat level is VL AND intervention urgency is VL THEN suppression effort 0.90, water drafting 0.80, containment line 0.70	When suppression feasibility is very high and the fire threat is very low and intervention urgency is very low, press the suppression effort at full strength (0.90), draft water hard (0.80), cut containment line hard (0.70).	Generative (Stage 3)	step 15, 2 min, deficit 0.05 gen_rule_0146
G24	IF intervention urgency is H AND suppression feasibility is L AND asset exposure risk is H THEN retardant drop 0.90, containment line 0.80, water drafting 0.70	When intervention urgency is high and suppression feasibility is low and asset exposure risk is high, drop retardant at full strength (0.90), cut containment line hard (0.80), draft water hard (0.70).	Generative (Stage 3)	step 10, 10 min, deficit 0.3 gen_rule_0154
G25	IF asset exposure risk is H AND evacuation pressure is VH AND suppression feasibility is L THEN water drafting 0.90, suppression effort 0.80, retardant drop 0.70	When asset exposure risk is high and evacuation pressure is very high and suppression feasibility is low, draft water at full strength (0.90), press the suppression effort hard (0.80), drop retardant hard (0.70).	Generative (Stage 3)	step 74, 74 min, deficit 0.05 gen_rule_0158
G26	IF intervention urgency is >=L AND suppression feasibility is M THEN suppression effort 0.90, containment line 0.80, retardant drop 0.80	When intervention urgency is at least low and suppression feasibility is moderate, press the suppression effort at full strength (0.90), cut containment line hard (0.80), drop retardant hard (0.80). Recorded reason: The VH-gated attack battery never fired; Agent_3 (614 fire cells) only reaches urgency ~0.36 and feasibility 0.42=M, so an M-level trigger is what actually orde	Generative (Stage 3)	gen_rule_0159
G27	IF asset exposure risk is >=M THEN retardant drop 0.80, containment line 0.70	When asset exposure risk is at least moderate, drop retardant hard (0.80), cut containment line hard (0.70). Recorded reason: Ground logistics in Agent_3 are 0.25; asset_exposure_risk reaches H(0.75)/M(0.60), and air-delivered retardant coats the head without needing road access.	Generative (Stage 3)	gen_rule_0160
G28	IF asset exposure risk is H AND	When asset exposure risk is high and	Generative (Stage 3)	step 60, 55 min, deficit 0.05

	suppression feasibility is H AND evacuation pressure is VH THEN suppression effort 0.90, water drafting 0.80, retardant drop 0.70	suppression feasibility is high and evacuation pressure is very high, press the suppression effort at full strength (0.90), draft water hard (0.80), drop retardant hard (0.70). Written 2 times over the recorded runs.		gen_rule_0165
G29	IF asset exposure risk is H AND suppression feasibility is H THEN suppression effort 0.90, water drafting 0.80, retardant drop 0.70	When asset exposure risk is high and suppression feasibility is high, press the suppression effort at full strength (0.90), draft water hard (0.80), drop retardant hard (0.70).	Generative (Stage 3)	step 60, 45 min, deficit 0.1 gen_rule_0167
G30	IF fire threat level is H AND suppression feasibility is VH AND intervention urgency is VL THEN suppression effort 1.00, water drafting 0.80, retardant drop 0.70	When the fire threat is high and suppression feasibility is very high and intervention urgency is very low, press the suppression effort at full strength (1.00), draft water hard (0.80), drop retardant hard (0.70).	Generative (Stage 3)	step 10, 0 min, deficit 0.0034173351935821716 gen_rule_0170
G31	IF fire threat level is VH AND suppression feasibility is VH AND asset exposure risk is M THEN suppression effort 1.00, water drafting 0.90, retardant drop 0.80	When the fire threat is very high and suppression feasibility is very high and asset exposure risk is moderate, press the suppression effort at full strength (1.00), draft water at full strength (0.90), drop retardant hard (0.80).	Generative (Stage 3)	step 116, 100 min, deficit 0.01834512224657575 gen_rule_0186
G32	IF suppression feasibility is VH AND asset exposure risk is H AND evacuation pressure is VH THEN suppression effort 1.00, water drafting 0.90, retardant drop 0.80	When suppression feasibility is very high and asset exposure risk is high and evacuation pressure is very high, press the suppression effort at full strength (1.00), draft water at full strength (0.90), drop retardant hard (0.80).	Generative (Stage 3)	step 138, 138 min, deficit 0.029920280860896376 gen_rule_0188
G33	IF suppression feasibility is VH AND asset exposure risk is VL THEN suppression effort 0.90, water drafting 0.80, containment line 0.70	When suppression feasibility is very high and asset exposure risk is very low, press the suppression effort at full strength (0.90), draft water hard (0.80), cut containment line hard (0.70).	Generative (Stage 3)	step 173, 168 min, deficit 0.042532150582870465 gen_rule_0189
G34	IF suppression feasibility is >=M AND evacuation pressure is H THEN containment line 0.90, retardant drop 0.80, resource deployment 0.70	When suppression feasibility is at least moderate and evacuation pressure is high, cut containment line at full strength (0.90), drop retardant hard (0.80), deploy resources hard (0.70).	Generative (Stage 3)	step 190, 180 min, deficit 0.0461273417120141 gen_rule_0191
G35	IF evacuation pressure is H AND asset exposure risk is VL THEN containment line 0.90, retardant drop 0.80, resource deployment 0.60	When evacuation pressure is high and asset exposure risk is very low, cut containment line at full strength (0.90), drop retardant hard (0.80), deploy resources moderately (0.60).	Generative (Stage 3)	step 41, 2460 min, deficit 0.2 gen_rule_0210
G36	IF fire threat level is VH AND suppression feasibility is VL AND evacuation pressure is VL THEN retardant drop 0.90, containment line 0.80, resource deployment 0.60	When the fire threat is very high and suppression feasibility is very low and evacuation pressure is very low, drop retardant at full strength (0.90), cut containment line hard (0.80), deploy resources moderately (0.60).	Generative (Stage 3)	step 15, 10 min, deficit 0.23661472179941942 gen_rule_0212
G37	IF asset exposure risk is VL AND evacuation pressure is VL THEN drafting retardant shuttle 0.90, containment line 0.70	When asset exposure risk is very low and evacuation pressure is very low, order drafting retardant shuttle at full strength (0.90), cut containment line	Generative (Stage 3)	step 39, 750 min, deficit 0.3 gen_rule_0214

		hard (0.70). "drafting retardant shuttle" is a generated intervention (retardant drop x1.0 + water drafting x0.6).		
G38	IF suppression feasibility is VL AND fire threat level is VL AND evacuation pressure is VL THEN retardant drop 0.80, containment line 0.70, water drafting 0.70	When suppression feasibility is very low and the fire threat is very low and evacuation pressure is very low, drop retardant hard (0.80), cut containment line hard (0.70), draft water hard (0.70).	Generative (Stage 3)	step 5, 270 min, deficit 0.3 gen_rule_0230
G39	IF asset exposure risk is VH AND evacuation pressure is VL AND intervention urgency is VL THEN containment line 0.80, retardant drop 0.80, asset protection 0.70	When asset exposure risk is very high and evacuation pressure is very low and intervention urgency is very low, cut containment line hard (0.80), drop retardant hard (0.80), protect the exposed assets hard (0.70).	Generative (Stage 3)	step 3, 0 min, deficit 0.0114670994675668 gen_rule_0235
G40	IF fire threat level is M AND asset exposure risk is VH AND suppression feasibility is VH THEN containment line 0.80, retardant drop 0.70, asset protection 0.70	When the fire threat is moderate and asset exposure risk is very high and suppression feasibility is very high, cut containment line hard (0.80), drop retardant hard (0.70), protect the exposed assets hard (0.70).	Generative (Stage 3)	step 5, 0 min, deficit 0.0032422616290203893 gen_rule_0249
G41	IF fire threat level is >=H AND suppression feasibility is VH AND intervention urgency is H THEN suppression effort 1.00, water drafting 0.90, retardant drop 0.80	When the fire threat is at least high and suppression feasibility is very high and intervention urgency is high, press the suppression effort at full strength (1.00), draft water at full strength (0.90), drop retardant hard (0.80).	Generative (Stage 3)	step 20, 510 min, deficit 0.3 gen_rule_0254
A1	IF asset exposure risk is VH AND suppression feasibility is VH THEN asset protection 0.88, evacuation 0.60	When asset exposure risk is very high and suppression feasibility is very high, protect the exposed assets at full strength (0.88), evacuate moderately (0.60). Instantiated from the standing rule A1 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 10, 10 min, deficit 0.18 evfis_mod_0002
A2	IF suppression feasibility is X1 AND asset exposure risk is H THEN asset protection 0.60, evacuation 0.51	When suppression feasibility is in the inserted band X1 and asset exposure risk is high, protect the exposed assets moderately (0.60), evacuate moderately (0.51). Instantiated from the standing rule A2 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 60, 60 min, deficit 0.010119975797321143 evfis_mod_0029
A3	IF suppression feasibility is VH AND asset exposure risk is H THEN asset protection 0.66, evacuation 0.58	When suppression feasibility is very high and asset exposure risk is high, protect the exposed assets hard (0.66), evacuate moderately (0.58). Instantiated from the standing rule A3 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 109, 109 min, deficit 0.05 evfis_mod_0044
A4	IF evacuation pressure is VH AND asset exposure risk is H THEN evacuation 0.63, asset protection 0.60	When evacuation pressure is very high and asset exposure risk is high, evacuate moderately (0.63), protect the exposed assets moderately (0.60). Instantiated from the standing rule A4	Resolution (Stage 2)	step 91, 91 min, deficit 0.1 evfis_mod_0047

		of the minimal base, on an antecedent cell that base did not answer.		
A5	IF evacuation pressure is X1 AND asset exposure risk is H THEN evacuation 0.63, asset protection 0.60	When evacuation pressure is in the inserted band X1 and asset exposure risk is high, evacuate moderately (0.63), protect the exposed assets moderately (0.60). Instantiated from the standing rule A5 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 96, 96 min, deficit 0.08 evfis_mod_0048
A6	IF asset exposure risk is VH AND suppression feasibility is M THEN asset protection 0.60, evacuation 0.32	When asset exposure risk is very high and suppression feasibility is moderate, protect the exposed assets moderately (0.60), evacuate lightly (0.32). Instantiated from the standing rule A6 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 0, 0 min, deficit 0.005000000000000001 evfis_mod_0084
A7	IF asset exposure risk is VH AND suppression feasibility is H THEN asset protection 0.75, evacuation 0.62	When asset exposure risk is very high and suppression feasibility is high, protect the exposed assets hard (0.75), evacuate moderately (0.62). Instantiated from the standing rule A7 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 5, 5 min, deficit 0.3 evfis_mod_0089
A8	IF asset exposure risk is M AND evacuation pressure is H THEN asset protection 0.53, evacuation 0.53	When asset exposure risk is moderate and evacuation pressure is high, protect the exposed assets moderately (0.53), evacuate moderately (0.53). Instantiated from the standing rule A8 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 5, 5 min, deficit 0.3 evfis_mod_0153
A9	IF suppression feasibility is H AND evacuation pressure is VH THEN evacuation 0.63, asset protection 0.60	When suppression feasibility is high and evacuation pressure is very high, evacuate moderately (0.63), protect the exposed assets moderately (0.60). Instantiated from the standing rule A9 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 35, 35 min, deficit 0.05 evfis_mod_0164
A10	IF suppression feasibility is VH AND asset exposure risk is M THEN asset protection 0.59, evacuation 0.51	When suppression feasibility is very high and asset exposure risk is moderate, protect the exposed assets moderately (0.59), evacuate moderately (0.51). Instantiated from the standing rule A10 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 5, 5 min, deficit 0.06 evfis_mod_0169
A11	IF suppression feasibility is X1 AND asset exposure risk is M THEN asset protection 0.59, evacuation 0.52	When suppression feasibility is in the inserted band X1 and asset exposure risk is moderate, protect the exposed assets moderately (0.59), evacuate moderately (0.52). Instantiated from the standing rule A11 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 10, 10 min, deficit 0.05 evfis_mod_0171
A12	IF evacuation pressure is VL	When evacuation pressure is very low	Resolution (Stage 2)	step 15, 15 min, deficit 0.3

	AND fire threat level is VL THEN evacuation 0.54, suppression effort 0.29, resource deployment 0.24	and the fire threat is very low, evacuate moderately (0.54), press the suppression effort lightly (0.29), deploy resources lightly (0.24). Instantiated from the standing rule A12 of the minimal base, on an antecedent cell that base did not answer.		evfis_mod_0175
A13	IF evacuation pressure is VL AND fire threat level is VH THEN evacuation 0.54, suppression effort 0.29, resource deployment 0.25	When evacuation pressure is very low and the fire threat is very high, evacuate moderately (0.54), press the suppression effort lightly (0.29), deploy resources lightly (0.25). Instantiated from the standing rule A13 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 25, 25 min, deficit 0.12 evfis_mod_0176
A14	IF evacuation pressure is X1 AND fire threat level is VH THEN evacuation 0.55, suppression effort 0.29, resource deployment 0.24	When evacuation pressure is in the inserted band X1 and the fire threat is very high, evacuate moderately (0.55), press the suppression effort lightly (0.29), deploy resources lightly (0.24). Instantiated from the standing rule A14 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 30, 30 min, deficit 0.06 evfis_mod_0177
A15	IF evacuation pressure is VH AND fire threat level is VH THEN evacuation 0.56, suppression effort 0.28, resource deployment 0.24	When evacuation pressure is very high and the fire threat is very high, evacuate moderately (0.56), press the suppression effort lightly (0.28), deploy resources lightly (0.24). Instantiated from the standing rule A15 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 42, 42 min, deficit 0.1 evfis_mod_0179
A16	IF suppression feasibility is X2 AND evacuation pressure is VL THEN evacuation 0.52, suppression effort 0.23	When suppression feasibility is in the inserted band X2 and evacuation pressure is very low, evacuate moderately (0.52), press the suppression effort lightly (0.23). Instantiated from the standing rule A16 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 5, 5 min, deficit 0.05 evfis_mod_0180
A17	IF evacuation pressure is X1 AND suppression feasibility is VH THEN evacuation 0.62, suppression effort 0.21	When evacuation pressure is in the inserted band X1 and suppression feasibility is very high, evacuate moderately (0.62), press the suppression effort lightly (0.21). Instantiated from the standing rule A17 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 241, 241 min, deficit 0.05 evfis_mod_0183
A18	IF suppression feasibility is X1 AND evacuation pressure is VH THEN evacuation 0.60, asset protection 0.54	When suppression feasibility is in the inserted band X1 and evacuation pressure is very high, evacuate moderately (0.60), protect the exposed	Resolution (Stage 2)	step 130, 130 min, deficit 0.06 evfis_mod_0187

		assets moderately (0.54). Instantiated from the standing rule A18 of the minimal base, on an antecedent cell that base did not answer.		
A19	IF suppression feasibility is X1 AND asset exposure risk is VH THEN asset protection 0.70, resource deployment 0.38	When suppression feasibility is in the inserted band X1 and asset exposure risk is very high, protect the exposed assets hard (0.70), deploy resources lightly (0.38). Instantiated from the standing rule A19 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 0, 0 min, deficit 0.00343502135448448 evfis_mod_0194
A20	IF asset exposure risk is X1 AND evacuation pressure is H THEN asset protection 0.60, evacuation 0.51	When asset exposure risk is in the inserted band X1 and evacuation pressure is high, protect the exposed assets moderately (0.60), evacuate moderately (0.51). Instantiated from the standing rule A20 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 38, 38 min, deficit 0.03691907450334458 evfis_mod_0197
A21	IF asset exposure risk is X1 AND evacuation pressure is VL THEN asset protection 0.70, evacuation 0.51	When asset exposure risk is in the inserted band X1 and evacuation pressure is very low, protect the exposed assets hard (0.70), evacuate moderately (0.51). Instantiated from the standing rule A21 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 5, 5 min, deficit 0.3 evfis_mod_0199
A22	IF suppression feasibility is H AND fire threat level is M THEN suppression effort 0.36, evacuation 0.30	When suppression feasibility is high and the fire threat is moderate, press the suppression effort lightly (0.36), evacuate lightly (0.30). Instantiated from the standing rule A22 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 5, 5 min, deficit 0.06 evfis_mod_0200
A23	IF fire threat level is VH AND evacuation pressure is H THEN suppression effort 0.57, evacuation 0.44, resource deployment 0.43	When the fire threat is very high and evacuation pressure is high, press the suppression effort moderately (0.57), evacuate moderately (0.44), deploy resources moderately (0.43). Instantiated from the standing rule A23 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 49, 2940 min, deficit 0.0016938060829069972 evfis_mod_0201
A24	IF fire threat level is X1 AND evacuation pressure is H THEN suppression effort 0.52, evacuation 0.44, resource deployment 0.39	When the fire threat is in the inserted band X1 and evacuation pressure is high, press the suppression effort moderately (0.52), evacuate moderately (0.44), deploy resources lightly (0.39). Instantiated from the standing rule A24 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 50, 3000 min, deficit 0.16 evfis_mod_0202
A25	IF evacuation pressure is X2 AND fire threat level is VH THEN evacuation 0.42, suppression effort 0.38	When evacuation pressure is in the inserted band X2 and the fire threat is very high, evacuate moderately (0.42),	Resolution (Stage 2)	step 53, 3180 min, deficit 0.1 evfis_mod_0204

		press the suppression effort lightly (0.38). Instantiated from the standing rule A25 of the minimal base, on an antecedent cell that base did not answer.		
A26	IF evacuation pressure is H AND suppression feasibility is VL THEN evacuation 0.40, suppression effort 0.27	When evacuation pressure is high and suppression feasibility is very low, evacuate moderately (0.40), press the suppression effort lightly (0.27). Instantiated from the standing rule A26 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 55, 3300 min, deficit 0.08 evfis_mod_0205
A27	IF evacuation pressure is X2 AND suppression feasibility is VL THEN evacuation 0.40, suppression effort 0.25	When evacuation pressure is in the inserted band X2 and suppression feasibility is very low, evacuate lightly (0.40), press the suppression effort lightly (0.25). Instantiated from the standing rule A27 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 56, 3360 min, deficit 0.12 evfis_mod_0206
A28	IF evacuation pressure is VL AND suppression feasibility is VL THEN evacuation 0.38, suppression effort 0.10	When evacuation pressure is very low and suppression feasibility is very low, evacuate lightly (0.38), press the suppression effort lightly (0.10). Instantiated from the standing rule A28 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 64, 3840 min, deficit 0.06 evfis_mod_0208
A29	IF asset exposure risk is H AND suppression feasibility is VL THEN asset protection 0.61, suppression effort 0.41	When asset exposure risk is high and suppression feasibility is very low, protect the exposed assets moderately (0.61), press the suppression effort moderately (0.41). Instantiated from the standing rule A29 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 0, 0 min, deficit 0.0026632334686656515 evfis_mod_0209
A30	IF fire threat level is VH AND intervention urgency is VH THEN suppression effort 0.51, resource deployment 0.49, evacuation 0.43	When the fire threat is very high and intervention urgency is very high, press the suppression effort moderately (0.51), deploy resources moderately (0.49), evacuate moderately (0.43). Instantiated from the standing rule A30 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 47, 1410 min, deficit 0.03356126899060469 evfis_mod_0215
A31	IF suppression feasibility is X3 AND evacuation pressure is VL THEN evacuation 0.42, suppression effort 0.37	When suppression feasibility is in the inserted band X3 and evacuation pressure is very low, evacuate moderately (0.42), press the suppression effort lightly (0.37). Instantiated from the standing rule A31 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 50, 1500 min, deficit 0.3 evfis_mod_0216
A32	IF suppression feasibility is X3	When suppression feasibility is in the	Resolution (Stage 2)	step 51, 1530 min, deficit 0.3

	AND evacuation pressure is VH THEN evacuation 0.42, suppression effort 0.33	inserted band X3 and evacuation pressure is very high, evacuate moderately (0.42), press the suppression effort lightly (0.33). Instantiated from the standing rule A32 of the minimal base, on an antecedent cell that base did not answer.		evfis_mod_0218
A33	IF evacuation pressure is X1 AND fire threat level is VL THEN evacuation 0.60, suppression effort 0.22, asset protection 0.17	When evacuation pressure is in the inserted band X1 and the fire threat is very low, evacuate moderately (0.60), press the suppression effort lightly (0.22), protect the exposed assets lightly (0.17). Instantiated from the standing rule A33 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 4, 120 min, deficit 0.3 evfis_mod_0223
A34	IF suppression feasibility is M AND evacuation pressure is VL THEN evacuation 0.29, asset protection 0.15	When suppression feasibility is moderate and evacuation pressure is very low, evacuate lightly (0.29), protect the exposed assets lightly (0.15). Instantiated from the standing rule A34 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 14, 420 min, deficit 0.3 evfis_mod_0231
A35	IF suppression feasibility is X2 AND evacuation pressure is VH THEN evacuation 0.51, asset protection 0.20	When suppression feasibility is in the inserted band X2 and evacuation pressure is very high, evacuate moderately (0.51), protect the exposed assets lightly (0.20). Instantiated from the standing rule A35 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 33, 990 min, deficit 0.006300812327565346 evfis_mod_0232
A36	IF suppression feasibility is VH AND evacuation pressure is M THEN evacuation 0.51, asset protection 0.21	When suppression feasibility is very high and evacuation pressure is moderate, evacuate moderately (0.51), protect the exposed assets lightly (0.21). Instantiated from the standing rule A36 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 37, 1110 min, deficit 0.005049667673106065 evfis_mod_0233
A37	IF suppression feasibility is X2 AND evacuation pressure is M THEN evacuation 0.51, asset protection 0.21	When suppression feasibility is in the inserted band X2 and evacuation pressure is moderate, evacuate moderately (0.51), protect the exposed assets lightly (0.21). Instantiated from the standing rule A37 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 38, 1140 min, deficit 0.005023316330322116 evfis_mod_0234
A38	IF suppression feasibility is VL AND asset exposure risk is M THEN asset protection 0.37, evacuation 0.37	When suppression feasibility is very low and asset exposure risk is moderate, protect the exposed assets lightly (0.37), evacuate lightly (0.37). Instantiated from the standing rule A38 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 75, 2250 min, deficit 0.3 evfis_mod_0251

A39	IF suppression feasibility is X3 AND asset exposure risk is M THEN asset protection 0.37, evacuation 0.37	When suppression feasibility is in the inserted band X3 and asset exposure risk is moderate, protect the exposed assets lightly (0.37), evacuate lightly (0.37). Instantiated from the standing rule A39 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 76, 2280 min, deficit 0.3 evfis_mod_0252
A40	IF suppression feasibility is H AND evacuation pressure is VL THEN evacuation 0.37, asset protection 0.36	When suppression feasibility is high and evacuation pressure is very low, evacuate lightly (0.37), protect the exposed assets lightly (0.36). Instantiated from the standing rule A40 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 37, 2220 min, deficit 0.3 evfis_mod_0255
A41	IF asset exposure risk is X1 AND suppression feasibility is L THEN asset protection 0.62, evacuation 0.45	When asset exposure risk is in the inserted band X1 and suppression feasibility is low, protect the exposed assets moderately (0.62), evacuate moderately (0.45). Instantiated from the standing rule A41 of the minimal base, on an antecedent cell that base did not answer.	Resolution (Stage 2)	step 87, 87 min, deficit 0.7072578114263436 evfis_mod_0257

Rules the evolving-fuzzy stage tuned

ID	Rule after tuning	Description	Steps	Produced
G5	IF asset exposure risk is VH AND suppression feasibility is VH THEN asset protection 1.00, downwind retardant shield 1.00, resource deployment 1.00	When asset exposure risk is very high and suppression feasibility is very high, protect the exposed assets at full strength (1.00), order downwind retardant shield at full strength (1.00), deploy resources at full strength (1.00). "downwind retardant shield" is a generated intervention (retardant drop x1.0 + containment line x0.8). The evolving-fuzzy stage moved this rule over 78 accepted steps: downwind retardant shield 0.90 -> 1.00 (+0.10); asset protection 0.70 -> 1.00 (+0.30); resource deployment 0.70 -> 1.00 (+0.30). A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.	78	step 12, 360 min, deficit 0.3 evfis_mod_0011 .. evfis_mod_0245
A5	IF evacuation pressure is X1 AND asset exposure risk is H THEN asset protection 0.35, evacuation 0.38	When evacuation pressure is in the inserted band X1 and asset exposure risk is high, protect the exposed assets lightly (0.35), evacuate lightly (0.38). The evolving-fuzzy stage moved this rule over 17 accepted steps: evacuation 0.63 -> 0.38 (-0.25); asset protection 0.60 -> 0.35 (-0.25). A step is only kept when the reseeded	17	step 47, 47 min, deficit 0.26 evfis_mod_0050 .. evfis_mod_0132

		forecast shows it costs less than the standing decision, so the walk is directional rather than a search.		
A7	IF asset exposure risk is VH AND suppression feasibility is H THEN asset protection 1.00, evacuation 1.00	When asset exposure risk is very high and suppression feasibility is high, protect the exposed assets at full strength (1.00), evacuate at full strength (1.00). The evolving-fuzzy stage moved this rule over 14 accepted steps: asset protection 0.75 -> 1.00 (+0.25); evacuation 0.62 -> 1.00 (+0.38). A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.	14	step 56, 56 min, deficit 0.3 evfis_mod_0091 .. evfis_mod_0104
A1	IF asset exposure risk is VH AND suppression feasibility is VH THEN asset protection 1.00, evacuation 1.00	When asset exposure risk is very high and suppression feasibility is very high, protect the exposed assets at full strength (1.00), evacuate at full strength (1.00). The evolving-fuzzy stage moved this rule over 13 accepted steps: asset protection 0.88 -> 1.00 (+0.12); evacuation 0.60 -> 1.00 (+0.40). A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.	13	step 29, 29 min, deficit 0.14 evfis_mod_0027 .. evfis_mod_0141
G27	IF asset exposure risk is >=M THEN containment line 0.95, retardant drop 1.00	When asset exposure risk is at least moderate, cut containment line at full strength (0.95), drop retardant at full strength (1.00). The evolving-fuzzy stage moved this rule over 13 accepted steps: retardant drop 0.80 -> 1.00 (+0.20); containment line 0.70 -> 0.95 (+0.25). A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.	13	step 0, 2580 min, deficit 0.3 evfis_mod_0173 .. evfis_mod_0256
G33	IF suppression feasibility is VH AND asset exposure risk is VL THEN containment line 0.60, suppression effort 0.80, water drafting 0.70	When suppression feasibility is very high and asset exposure risk is very low, cut containment line moderately (0.60), press the suppression effort hard (0.80), draft water hard (0.70). The evolving-fuzzy stage moved this rule over 9 accepted steps: suppression effort 0.90 -> 0.80 (-0.10); water drafting 0.80 -> 0.70 (-0.10); containment line 0.70 -> 0.60 (-0.10). A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.	9	step 16, 480 min, deficit 0.3 evfis_mod_0236 .. evfis_mod_0248
A6	IF asset exposure risk is VH AND suppression feasibility is M THEN asset protection 0.65, evacuation 0.37	When asset exposure risk is very high and suppression feasibility is moderate, protect the exposed assets hard (0.65), evacuate lightly (0.37). The evolving-fuzzy stage moved this	3	step 47, 47 min, deficit 0.3 evfis_mod_0085 .. evfis_mod_0087

		rule over 3 accepted steps: asset protection 0.60 -> 0.65 (+0.05); evacuation 0.32 -> 0.37 (+0.05). A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.		
A19	IF suppression feasibility is X1 AND asset exposure risk is VH THEN asset protection 0.85, resource deployment 0.53	When suppression feasibility is in the inserted band X1 and asset exposure risk is very high, protect the exposed assets hard (0.85), deploy resources moderately (0.53). The evolving-fuzzy stage moved this rule over 3 accepted steps: asset protection 0.70 -> 0.85 (+0.15); resource deployment 0.38 -> 0.53 (+0.15). A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.	3	step 58, 1740 min, deficit 0.3 evfis_mod_0195 .. evfis_mod_0220
R26	IF fire threat level is VH THEN containment line 0.80, resource deployment 0.90, retardant drop 0.95, suppression effort 1.00	When the fire threat is very high, cut containment line hard (0.80), deploy resources at full strength (0.90), drop retardant at full strength (0.95), press the suppression effort at full strength (1.00). The evolving-fuzzy stage moved this rule over 2 accepted steps: resource deployment 0.80 -> 0.90 (+0.10). Unchanged: suppression effort, retardant drop, containment line. A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.	2	evfis_mod_0134 .. evfis_mod_0136
A17	IF evacuation pressure is X1 AND suppression feasibility is VH THEN evacuation 0.62, suppression effort 0.21	When evacuation pressure is in the inserted band X1 and suppression feasibility is very high, evacuate moderately (0.62), press the suppression effort lightly (0.21). The evolving-fuzzy stage moved this rule over 2 accepted steps: no consequent ended away from where it started. Unchanged: evacuation, suppression effort. A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.	2	step 12, 10 min, deficit 0.05 evfis_mod_0185 .. evfis_mod_0196
A12	IF evacuation pressure is VL AND fire threat level is VL THEN evacuation 0.44, resource deployment 0.14, suppression effort 0.19	When evacuation pressure is very low and the fire threat is very low, evacuate moderately (0.44), deploy resources lightly (0.14), press the suppression effort lightly (0.19). The evolving-fuzzy stage moved this rule over 2 accepted steps: evacuation 0.54 -> 0.44 (-0.10); suppression effort 0.29 -> 0.19 (-0.10); resource deployment 0.24 -> 0.14 (-0.10). A step is only kept	2	step 0, 0 min, deficit 0.00244130726675554 evfis_mod_0250 .. evfis_mod_0253

		when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.		
G18	IF suppression feasibility is M AND asset exposure risk is VH AND evacuation pressure is VH THEN retardant drop 0.75, suppression effort 0.95, water drafting 0.85	When suppression feasibility is moderate and asset exposure risk is very high and evacuation pressure is very high, drop retardant hard (0.75), press the suppression effort at full strength (0.95), draft water at full strength (0.85). The evolving-fuzzy stage moved this rule over 1 accepted steps: suppression effort 0.90 -> 0.95 (+0.05); water drafting 0.80 -> 0.85 (+0.05); retardant drop 0.70 -> 0.75 (+0.05). A step is only kept when the reseeded forecast shows it costs less than the standing decision, so the walk is directional rather than a search.	1	step 11, 11 min, deficit 0.3 evfis_mod_0090 .. evfis_mod_0090

Input variables the system reshaped

Variable	Change	New terms	Why	Produced
(unnamed variable)	moved a shared boundary of the partition	X1 [0.67, 0.74, 0.84, 0.91]	the boundary between two terms sat where the decision needed to distinguish; both sides move together, so the partition stays intact	step 60, 60 min, deficit 0.010119975797321143 evfis_mod_0030
(unnamed variable)	moved a shared boundary of the partition	-	the boundary between two terms sat where the decision needed to distinguish; both sides move together, so the partition stays intact	step 3, 3 min, deficit 0.12 evfis_mod_0042
(unnamed variable)	moved a shared boundary of the partition	X1 [0.51, 0.58, 0.68, 0.75]	the boundary between two terms sat where the decision needed to distinguish; both sides move together, so the partition stays intact	step 96, 96 min, deficit 0.08 evfis_mod_0049
(unnamed variable)	inserted the term X1, X2 into the partition	X1 [0.67, 0.74, 0.84, 0.91], X2 [0.47, 0.54, 0.64, 0.71]	the five standing terms were too coarse to separate the situation the rule was facing, so a term was added between them; the partition still sums to one everywhere, so no existing rule changed meaning outside the new band	step 5, 5 min, deficit 0.05 evfis_mod_0181
(unnamed variable)	inserted the term X1 into the partition	X1 [0.48, 0.55, 0.65, 0.72]	the five standing terms were too coarse to separate the situation the rule was facing, so a term was added between them; the partition still sums to one everywhere, so no existing rule changed meaning outside the new band	step 38, 38 min, deficit 0.03691907450334458 evfis_mod_0198
(unnamed variable)	moved a shared boundary of the partition	X1 [0.40, 0.47, 0.57, 0.64]	the boundary between two terms sat where the decision needed to distinguish; both sides move together, so the partition stays intact	step 50, 3000 min, deficit 0.16 evfis_mod_0203
(unnamed variable)	inserted the term X1, X2 into the partition	X1 [0.51, 0.58, 0.68, 0.75], X2 [0.28, 0.35, 0.45, 0.52]	the five standing terms were too coarse to separate the situation the rule was facing, so a term was added	step 56, 3360 min, deficit 0.12 evfis_mod_0207

			between them; the partition still sums to one everywhere, so no existing rule changed meaning outside the new band	
(unnamed variable)	inserted the term X1, X2, X3 into the partition	X1 [0.67, 0.74, 0.84, 0.91], X2 [0.47, 0.54, 0.64, 0.71], X3 [0.30, 0.37, 0.47, 0.54]	the five standing terms were too coarse to separate the situation the rule was facing, so a term was added between them; the partition still sums to one everywhere, so no existing rule changed meaning outside the new band	step 50, 1500 min, deficit 0.3 evfis_mod_0217
(unnamed variable)	moved a shared boundary of the partition	X1 [0.67, 0.74, 0.84, 0.91], X2 [0.47, 0.54, 0.64, 0.71], X3 [0.30, 0.37, 0.47, 0.54]	the boundary between two terms sat where the decision needed to distinguish; both sides move together, so the partition stays intact	step 3, 90 min, deficit 0.3 evfis_mod_0237